Asymmetries in Spontaneous and Instructed Deception

Josiah Luikham JOSIAH.LUIKHAM@GMAIL.COM
Independent Researcher

arXiv:2609.00180v1 [cs.AI] 31 Aug 2026

Abstract

Large language models sometimes deceive users without being instructed to. However, much of the study on deception in models involves instructed deception. We investigated the relationship between instructed and spontaneous (uninstructed) deception in Llama-3.1-70B-Instruct. We compared these two deception settings through direction geometry, cross-setting classifiers, and cross-setting steering. We found the two deception settings share a component of direction (cosine of approximately 0.5) and an asymmetry in the transfer between settings regarding detection and causation. Spontaneous trained classifiers performed better on instructed data than vice versa, and instructed derived directions performed better at steering spontaneous prompts than vice versa. Likewise the best token position to derive steering vectors from differed from the best token position to train and apply classifiers.

1. Introduction

Large language models sometimes produce deceptive output when interacting with users (Park et al., 2023). This raises many practical and safety concerns for the use of LLMs in serious settings. Understanding the nature of deception in LLMs and how it may be mitigated is therefore a matter of great importance.

In a deployed environment, when an LLM deceives, it deceives spontaneously without an instruction to do so. Therefore the study of spontaneous deception is most relevant to real world concerns. However, much existing work is focused on instructed deception. Some prior work has studied the relationship between instructed and spontaneous deception. Goldowsky-Dill et al. (2025) studied the generalization of probes trained on instructed deception to spontaneous deception. Following this, we investigated the full symmetry between spontaneous and instructed deception in Llama-3.1-70B-Instruct (Grattafiori et al., 2024). We tested how well classifiers trained on both instructed and spontaneous deception generalized to the other setting. Moreover, we explored whether the relationship between spontaneous and instructed deception is only correlational or if it is causal as well through direction steering. We also considered type level deception (fabrication and omission) instead of just binary deception/honesty.

We created datasets of instructed and spontaneous deception prompts and model responses. Using these, we investigated the transfer of classifiers trained and applied between settings, the transfer of directional steering between settings, and the geometry of the setting deception directions compared with each other.

We found the following results:

- **A shared deception component across settings.** The setting deception directions computed from the model's response tokens shared a cosine of approximately 0.5 compared to chance 0.011.

© J. Luikham.

- **Detection transfers asymmetrically.** We found that probes trained on spontaneous deception achieve near ceiling accuracy when applied to instructed deception. Instructed deception trained probes did not perform as well between settings.
- **Causal Directions transfer, but asymmetrically.** We found that steering one setting's prompts with directions derived from the other setting does suppress and induce deception, although instructed directions perform best.
- **Read in, Write out token asymmetry.** We found a difference in the most effective token positions for probe training and for deriving steering directions. Response tokens generally performed best both for training and applying probes, but the last prompt token was the best place to derive steering vectors from.

2. Related Work

Generalization of deception probes Goldowsky-Dill et al. (2025) trained probes on the internal states of models instructed to be deceptive or honest and tested those probes in scenarios where the model would deceived spontaneously (insider trading etc.). We trained probes on instructed and spontaneous deception activations, and tested the generalization between the two settings. We also probed and steered for type level deception (fabrication, omission) in our experiments.

Deception type representations Natarajan et al. (2026) showed that different types of deception can have different representations in a model, although some forms of deception overlap heavily. We also investigate type level deception, specifically fabrication and omission.

Causation vs detection Galeone et al. (2026) found that the direction which represented a model's recognition/knowledge of a subject was orthogonal to its refusal to answer questions regarding that subject. We also found dissociations between detection and control for deception.

3. Setup and methods

We generated datasets of prompts to test instructed and spontaneous deception. All System Prompts contained a scenario and a ground truth fact and all user prompts asked a question related to the ground truth. The instructed deception prompts contained an instruction to behave honestly or deceptively(according to omission or fabrication). The spontaneous deception prompts included an incentive to deceive in the scenario, split into omission and fabrication elicitation lists. Our definitions of fabrication and omission and objects/audiences of deception in scenarios followed the taxonomy of Shi et al. (2026). The prompts were used to generate responses from the model and these were appended and saved for downstream experiments. Spontaneous responses were divided into deceptive and honest lists based on assessment of their content by an LLM judge.

We trained logistic regression probes (Alain and Bengio, 2018) on the teacher forced activations of the prompt and response datasets to distinguish honest responses from deceptive responses binarily as well as by deception type (fabrication, omission). The probes were trained over various layers at the last prompt token or response tokens. They were

then applied to the teacher forced activations of a held out portion of the datasets at the same layers and last prompt token or response tokens.

We computed mass mean difference directions (Li et al., 2024) for deception from the spontaneous and instructed prompt/response datasets and used them to generate new responses from the prompts. Directions were derived from the last prompt token and the response tokens over layers 19-32. The normalized directions were then applied with coefficients of up to |1.5| at layers 19-32 over all tokens at generation time. We also performed response generation with random direction steering as a control.

Relatively few deceptive fabrication responses were elicited at baseline(without steering) with prompts in the spontaneous dataset, so we implemented additional tests to measure suppression steering effectiveness in this case. Of prompts which produced fabricative responses at baseline, we recorded which produced honest responses under fabrication suppression steering and under random steering. Flip rates from fabricative to honest were recorded with Wilson 95 percent intervals (Wilson, 1927) and the difference between the real and random-direction flip rates were recorded with Newcombe 95 percent intervals built from the Wilson intervals (Newcombe, 1998).

Response labeling and response deception ratings were performed by a GPT-4.1 Mini Judge. Spontaneous Prompt/Response pairs used for direction extraction and probe training/application were labeled as honest, fabricative, or omissive through a judgement/filtration pipeline (Appendix A). Instructed Prompt/Response pairs used for direction extraction and probe training/application were labeled based on their system prompt instruction. Steered and baseline responses of both datasets were scored on a 0-100 scale for fabrication and omission as well as coherence.

To validate the LLM Judge, we hand labeled 120 responses using only the resources available to the judge during its rating. The samples were stratified to contain diverse deception scores and steering levels. We found the human rater and LLM Judge agreed reasonably well. Binary fabrication ratings(responses rated with trait ≥ 50) agreed 85 percent of the time. Binary omission ratings agreed 82 percent of the time. Full documentation of the judge validation is available in (Appendix B).

4. Results

Train → eval	Balanced acc.	Retention	AUROC
instructed → instructed (response)	**1.000**	*(ceiling)*	1.000
spontaneous → instructed (response)	0.980	0.980	0.999 ± 0.000
spontaneous → spontaneous (response)	**0.934 ± 0.019**	*(ceiling)*	0.981 ± 0.008
instructed (prompt) → spontaneous (response)	0.893 ± 0.021	0.956	0.918 ± 0.024
spontaneous → spontaneous (prompt)	**0.530 ± 0.009**	*(ceiling)*	0.604 ± 0.005

Table 1: Binary (honest vs. deceptive) probe results at layer 35. Values are mean ± half-range over seeds 42,43, and 44. Bold rows are within-setting diagonals. Retention refers to the off-diagonal balanced accuracy divided by the diagonal of the same eval set.

We found that binary and type level deception probes performed best at layer 35 and succeeded asymmetrically in transfer between datasets. Probes trained on spontaneous response token activations and applied over instructed response token activations performed the best of our between setting probes, achieving near ceiling accuracy. Instructed trained binary probes applied to the spontaneous setting achieved their best scores when trained at the last prompt token and applied over the response tokens, but failed to match their spontaneous trained counterparts. Instructed trained type level probes applied to the spontaneous setting failed to achieve better than chance. Interestingly, we found that spontaneous activation trained probes applied within setting at the last prompt token failed to achieve good scores, indicating a lack of spontaneous deception signal at this point. Binary probe transfer results are given in Table 1. Type level probe results are given in Table 2.

	Per-class accuracy		
Train → eval	Fabrication	Omission	Honest
instructed → instructed	1.000	1.000	1.000
spontaneous → instructed	1.000	0.920	0.940
spontaneous → spontaneous	0.925	0.771 ± 0.052	0.945 ± 0.007
instructed → spontaneous	0.164	0.393 ± 0.022	0.995 ± 0.007

Table 2: Type-level (fabrication/omission/honest) classification results at layer 35 for response trained probes applied at response tokens. Values are mean ± half-range over seed 42, 43, and 44. Bold rows are within-setting diagonals. Retention refers to the off-diagonal balanced accuracy divided by the diagonal of the same eval set.

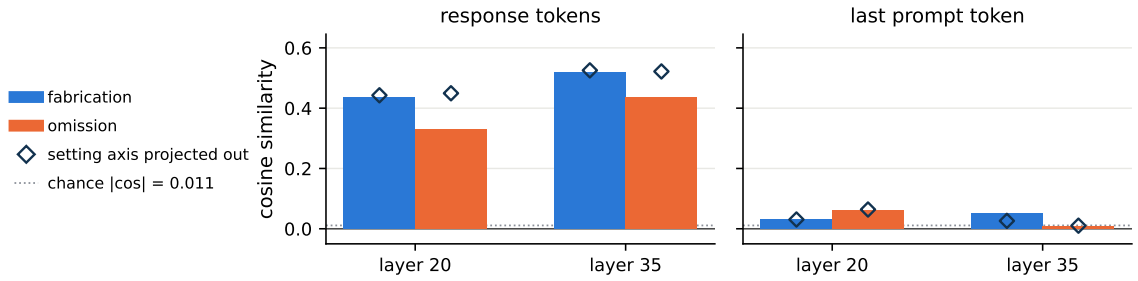


Figure 1: The cosine similarity between deception directions derived from activations in the spontaneous and instructed scenarios.

To determine the alignment of the spontaneous and instructed deception directions, we computed their cosine. The charted geometry can be seen in Figure 1. The cosine between the setting deception directions at layer 35 over the response tokens was 0.519 and 0.435 for fabrication and omission respectively versus chance 0.011. When the setting direction was projected out of the deception directions their cosine increased. The cosine between